

CENG381 Stochastic Processes

Rate of Convergence — Practice 1 (Answer Key)

Q1 Solution

a) Stationary distribution ($p=0.4, q=0.5$):

$$\begin{aligned}\pi_0 &= q/(p+q) = 0.5/(0.4+0.5) = 0.5/0.9 = 5/9 \approx 0.556 \\ \pi_1 &= p/(p+q) = 0.4/0.9 = 4/9 \approx 0.444\end{aligned}$$

b) Convergence derivation:

$$\begin{aligned}\mu_{\{t+1\}}(0) &= \mu_t(0) \cdot (1-p) + \mu_t(1) \cdot q \\ &= \mu_t(0) \cdot (1-p) + (1-\mu_t(0)) \cdot q \\ &= \mu_t(0) \cdot (1-p-q) + q\end{aligned}$$

Subtract $\pi_0 = q/(p+q)$ from both sides:

$$\begin{aligned}\Delta_{\{t+1\}} &= \mu_{\{t+1\}}(0) - \pi_0 \\ &= \mu_t(0) \cdot (1-p-q) + q - q/(p+q) \\ &= (1-p-q) \cdot (\mu_t(0) - q/(p+q)) \\ &= (1-p-q) \cdot \Delta_t \\ \lambda_2 &= 1 - p - q = 1 - 0.4 - 0.5 = 0.1\end{aligned}$$

c) Steps needed for $|\Delta_t| < 0.01$:

$$\begin{aligned}\Delta_0 &= \mu_0(0) - \pi_0 = 0 - 5/9 = -5/9 \approx -0.556 \\ |\Delta_t| &= |\lambda_2|^t \cdot |\Delta_0| = (0.1)^t \cdot (5/9)\end{aligned}$$

Solve $(0.1)^t \cdot (5/9) < 0.01$:

$$(0.1)^t < 0.01 \times 9/5 = 0.018$$

$$t \cdot \ln(0.1) < \ln(0.018)$$

$$t > \ln(0.018)/\ln(0.1) = (-4.017)/(-2.303) \approx 1.74$$

Minimum $t = 2$ steps.

$$\text{Check: } (0.1)^2 \cdot (5/9) = 0.01 \cdot 0.556 \approx 0.0056 < 0.01 \quad \checkmark$$

The chain converges extremely fast because $\lambda_2 = 0.1$ is very close to 0.

Q2 Solution

a) Eigenvalues of P :

$$\lambda_1 = 1 \quad (\text{every stochastic matrix has 1 as an eigenvalue})$$

CENG381 Stochastic Processes

Rate of Convergence — Practice 1 (Answer Key)

$$\lambda_2 = 1 - p - q = 1 - 0.4 - 0.5 = 0.1$$

b) Derivation of $p_{\{00\}}^{(n)}$:

Decompose P into the rank-1 part (projection onto stationary dist) plus a residual:

$$P = \Pi + \lambda_2 \cdot (P - \Pi)$$

where Π has all rows equal to $\pi = (5/9, 4/9)$. The rank-1 matrix Π satisfies $\Pi^2 = \Pi$ (idempotent), and $(P - \Pi)^2 = \lambda_2 \cdot (P - \Pi)$, so:

$$P^n = \Pi + \lambda_2^n \cdot (P - \Pi)$$

$$\begin{aligned} \text{Entry } (0,0): \quad p_{\{00\}}^{(n)} &= \pi_0 + \lambda_2^n \cdot (P_{\{00\}} - \pi_0) \\ &= 5/9 + (0.1)^n \cdot (0.6 - 5/9) \\ &= 5/9 + (0.1)^n \cdot (0.6 - 0.556) \\ &= 5/9 + (0.1)^n \cdot (4/9 \cdot \dots) \end{aligned}$$

More directly, using the standard formula for 2-state chains:

$$p_{\{00\}}^{(n)} = \pi_0 + \lambda_2^n \cdot \pi_1 = 5/9 + (0.1)^n \cdot (4/9)$$

c) Compute $p_{\{00\}}^{(5)}$ and the limit:

$$\begin{aligned} p_{\{00\}}^{(5)} &= 5/9 + (0.1)^5 \cdot (4/9) \\ &= 5/9 + 0.00001 \cdot (4/9) \\ &\approx 0.5556 + 0.0000044 \\ &\approx 0.5556 \quad (\text{chain has essentially converged!}) \end{aligned}$$

$$\lim_{n \rightarrow \infty} p_{\{00\}}^{(n)} = 5/9 = \pi_0 \quad \checkmark$$

Q3 Solution

a) Second eigenvalue and convergence speed:

$$\text{Server A: } \lambda_2(A) = 1 - 0.2 - 0.3 = 0.5$$

$$\text{Server B: } \lambda_2(B) = 1 - 0.1 - 0.05 = 0.85$$

Server A converges FASTER because $|\lambda_2(A)| = 0.5 < 0.85 = |\lambda_2(B)|$. A smaller $|\lambda_2|$ means the deviation from stationarity shrinks more rapidly each step. Server B has very sticky states (high self-loop

CENG381 Stochastic Processes

Rate of Convergence — Practice 1 (Answer Key)

probabilities 0.9 and 0.95), so it mixes slowly.

b) Stationary distributions:

Server A: $\pi_0(A) = q/(p+q) = 0.3/0.5 = 0.6$, $\pi_1(A) = 0.4$
Server B: $\pi_0(B) = q/(p+q) = 0.05/0.15 = 1/3 \approx 0.333$, $\pi_1(B) = 2/3$

Server A has higher long-run availability: 60% vs. 33.3% for Server B. Server B fails more often relative to its repair rate.

c) Steps for $|p_{\{00\}}^{(n)} - \pi_0| < 0.001$:

Using $|p_{\{00\}}^{(n)} - \pi_0| = |\lambda_2|^n \cdot \pi_1$:

Server A: $(0.5)^n \cdot 0.4 < 0.001$
 $(0.5)^n < 0.0025$
 $n > \log(0.0025)/\log(0.5) = 8.64 \rightarrow n \geq 9 \text{ steps}$

Server B: $(0.85)^n \cdot (2/3) < 0.001$
 $(0.85)^n < 0.0015$
 $n > \log(0.0015)/\log(0.85) = 39.9 \rightarrow n \geq 40 \text{ steps}$

Server A converges in 9 steps; Server B needs 40 steps — over 4× longer. This illustrates that even a modest increase in λ_2 (from 0.5 to 0.85) can dramatically slow down the mixing of the chain.